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SECP3106 - APPLICATION DEVELOPMENT

(WBL)

SECTION 02

**LAB ASSIGNMENT 2:
INDUSTRY PROJECT MODULE PROPOSAL**

INDUSTRY:

TADIKA AQIL MIQAIL (GENIUS AULAD SKUDAI)

LECTURER:

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Project Title

The title of the current project is Teacher Management System (GENIUSAQIL OS).

Introduction

Tadika Aqil Miqail, operating under the Genius Aulad franchise, is a well-established Islamic-based kindergarten network with a total of 91 branches nationwide. This project is specifically focused on the Genius Aulad Skudai branch, which currently accommodates approximately 9 teaching and administrative staff members serving a growing student population. As the branch continues to expand, the need for a structured and efficient management system has become increasingly evident.

Project Background

Dr. Norhanisah Abdullah, the Principal of Genius Aulad Skudai, highlighted that the school currently has no dedicated system to support the management of its teaching staff. All administrative processes including maintaining teacher records, assigning daily duties, handling leave applications, tracking training activities, and conducting performance evaluations are carried out entirely by hand, relying on paper forms, handwritten notes, and personal memory.

This fragmented approach has created significant operational difficulties for example records are hard to retrieve, leave requests lack a proper audit trail, task assignments are inconsistent, and there is little visibility over teacher development and performance. As the school grows, the manual workload has become increasingly unmanageable, raising the risk of human error and reducing transparency and accountability across daily operations.

In response to these challenges, this project aims to develop GeniusAqilOS, an Android-based Teacher Management System that centralises and digitises all teacher-related administrative processes into one integrated platform. The proposed system consists of six core industry modules: Teacher Record Management, Leave Management, Task and Duty Manager, Teacher Training Tracker, Teacher Performance Tracker, and Teacher Reporting System. Each module is designed to automate routine tasks, reduce manual effort, and provide the Principal with real-time oversight of school operations. The development of this system is justified not only by the urgent need to reduce administrative burden and human error, but also by its broader impact into a well-managed teaching workforce directly

supports better classroom performance and educational outcomes, making GeniusAqilOS a strategic investment in the long-term quality and growth of Genius Aulad Skudai.

Objectives

1. To develop a centralised Android-based Teacher Management System (GeniusAqilOS) that digitises and automates all teacher-related administrative processes at Genius Aulad Skudai.
2. To improve operational efficiency and data accuracy by replacing manual, paper-based workflows with structured digital modules covering teacher records, leave management, task assignment, training tracking, performance evaluation, and issue reporting.
3. To enhance administrative transparency and accountability by providing the Principal with real-time oversight, automated report generation, and a reliable audit trail across all staff management activities.

Target Users

The target users for the GENIUSAQILOS Teacher Management System are:

1. **Principal / Admin**

The Principal or Admin is responsible for managing the entire system. They will handle teacher records, approve or reject leave applications, assign and monitor tasks and duties, track teacher training and performance, and review reports submitted by teachers. They also ensure smooth school operations through centralized system control.

2. **Teachers / Staff**

There are around 9 teachers and staff who will use the system to update their personal information, apply for leave, check assigned duties, submit training details, view performance evaluations, and report any issues such as maintenance problems or workplace concerns. The system helps them manage their daily responsibilities more efficiently.

Mandatory Modules

The mandatory baseline modules for this project are Login, Registration, and Manage User. These modules are compulsory for every team and must be included together with the six industry modules.

Mandatory Module	Description	Feature Example
Login	Allows authorised users such as admin, teachers, and management to securely access the system using their registered credentials.	• Email & password authentication • Role-based access control (Admin / Teacher) • Session management & auto-logout • Password reset via OTP • eKYC identity verification for first-time login
Registration	Allows new users to create an account with basic information and validation before accessing the system.	• New account creation form • Field validation (email format, password strength) • Email verification upon registration • eKYC document upload for identity confirmation • Admin approval workflow before account activation
Manage User	Allows the admin to view, edit, remove, and manage user accounts based on their roles, such as teacher, admin, or management.	• View all registered users with role filter • Edit user profile and role assignment • Activate / deactivate user accounts • Reset user passwords • Audit log of user account changes

Six Industry Modules

The industry module for this project includes 6 modules: Teacher Records, Leave Management, Task & Duty Manager, Teacher Training Tracker, Teacher Performance Tracker, and Teacher Reporting System. These modules are the core for the whole project,

Industry Modules	Description	Feature example
Teacher Records	This module allows teachers to upload, view, access, edit and update their important information or documents like id, birth certificate and so on. They are also able to track their document status	• Secure document upload (IC, certs, contracts) • Document status tracker (Pending / Verified / Rejected) • Edit and update personal particulars • Admin verification and approval workflow • Document expiry alerts and renewal reminders
Leave Management	This module allows the teachers to apply leave and track their leave balance. Then, the approval and rejection will notify the teacher by principal. The leave data will be recorded. These applications can be appealed to the principal.	• Leave application form with type selection (MC, Annual, Emergency) • Real-time leave balance display • Push notification on approval or rejection • Appeal submission for rejected leaves • Leave history and audit trail for Admin
Task & Duty Manager	This module allows management to assign and manage teachers' daily and scheduled duties. It generates fair schedules based on availability, provides task checklist, tracks completion and send reminders to ensure responsibilities are carried out properly	• Duty assignment by date, time, and teacher • Auto-scheduler based on teacher availability • Task checklist with completion tracking • Push reminders for upcoming duties • Admin dashboard for duty overview
Teacher Training Tracker	This module records all teacher training activities, including title, date, and location, while monitoring each	• Training log entry (title, date, venue, hours) • Minimum training hours target tracker

	teacher's progress against a minimum training target. Generates yearly reports to ensure continuous professional development.	• Certificate upload and storage • Progress bar per teacher • Auto-generated yearly training reports
Teacher Performance Tracker	This module evaluates teacher performance through a customisable KPI checklist with auto-calculated scores, providing a clear overview of each teacher's overall performance. Also stores warning letters and misconduct records, with yearly reports generated for review.	• Customisable KPI checklist (punctuality, teaching quality, etc.) • Auto-calculated performance scores • Warning letter issuance and storage • Misconduct record management • Yearly performance report generation
Teacher Reporting System	This module allows teachers to easily submit reports on issues such as damage, bullying, harassment, or complaints, with the option for anonymous reporting and photo uploads. A built-in notification system keeps all parties updated whenever action is taken on a report.	• Report submission form (category: damage, complaint, misconduct) • Optional anonymous reporting toggle • Photo/evidence upload • Status tracker (Submitted / In Review / Resolved) • Push notification on every status update to relevant parties

Module Distribution Table

Each 9 modules will be distributed equally for each member based on their fundamental strength and weakness. // make/refine this sentence more wow

Module Type	Modules Name	Lead Member
Mandatory	Login	Dheshieghan
	Registration	Dheshieghan
	Manage User	Pravinraj
Industry	Teacher Records	Ziyaad
	Leave Management	Dheshieghan
	Task & Duty Manager	Pravinraj
	Teacher Training Tracker	Afiq
	Teacher Performance Tracker	Afiq & Adib
	Teacher Reporting System	Adib

Justification

The distribution of modules among team members was carefully planned based on each member's workload capacity, technical strengths, and the logical grouping of related functionalities.

Dheshieghan was assigned the Login and Leave Management modules because both involve authentication logic, user session handling, and form-based interactions, making it natural to group them under one developer who already manages system access. Pravinraj handles both Manage User and Task & Duty Manager, as both modules revolve around administrative control by managing user roles and overseeing teacher responsibilities are closely related operations that benefit from a consistent development approach.

Afiq was assigned the Teacher Training Tracker and co-leads the Teacher Performance Tracker, since both modules deal with monitoring teachers' professional growth and generating yearly reports, requiring similar data structures and reporting logic. Adib co-leads the Teacher Performance Tracker and solely leads the Teacher Reporting System, as both involve record-keeping of sensitive matters such as misconduct and complaints, ensuring consistent handling of confidential data. Ziyaad manages the Teacher Records module independently, as it serves as the foundational data source referenced by other modules, requiring focused attention to document management and data integrity.

Overall, this distribution ensures a balanced workload, minimises cross-dependency conflicts, and allows each member to develop deeper expertise in their assigned area while maintaining cohesion across the entire system.

Conclusion

The proposed GeniusAqilOS (Teacher Management System) aims to fundamentally transform the administrative landscape of Tadika Aqil Miqail, Genius Aulad Skudai by replacing fragmented manual processes with a unified, efficient digital platform. Spanning nine modules across mandatory and industry categories which namely Login, Registration, Manage User, Teacher Records, Leave Management, Task & Duty Manager, Teacher Training Tracker, Teacher Performance Tracker, and Teacher Reporting System. The system collectively addresses every key pain point identified during the needs assessment.

Through the centralisation of data and automation of routine tasks, GeniusAqilOS will equip the Principal with comprehensive, real-time oversight from monitoring daily duty schedules and processing leave applications to evaluating staff performance and managing training compliance. This empowers the school leadership to make informed decisions swiftly and confidently, while significantly reducing the administrative burden on both the Principal and teaching staff.

Beyond operational efficiency, the system promotes a culture of accountability, transparency, and continuous professional development pillars that are essential for sustaining a high-quality early childhood education environment. As Genius Aulad Skudai continues to grow, GeniusAqilOS represents a forward-looking, scalable investment that will directly support better teaching quality and, ultimately, improved outcomes for the children in its care.

Estimated Project Expenses

The following table provides an estimated breakdown of costs associated with the development and deployment of GeniusAqilOS.

No.	Expense Item	Description	Estimated Cost (MYR)
1	Flutter SDK & Dart	Open-source, free to use	Free
2	Firebase / Backend Services	Firestore, Auth, Storage, Cloud Messaging (Spark/Blaze Plan)	Free ~ RM 50/month

3	Google Play Developer Account	One-time registration fee for publishing the app	RM 105.00
4	UI/UX Design Tools	Figma Pro for wireframing and prototyping (1 month)	RM 85.00
5	FlutterFlow (<i>optional</i>)	Visual Flutter builder for rapid prototyping	Free ~ RM 130/month
6	VS Code / Android Studio	IDEs for Flutter development	Free
7	Testing Devices	Physical Android devices for UAT (assumed available)	RM 0.00
8	Documentation & Printing	Printing of proposal, reports, and user manuals	RM 60.00
		TOTAL (Estimated)	RM 250.00 ~ RM 430.00